

The Laws of Civilizational Coordination: A Unified Theory from Game Theory, Information Theory, and Statistical Mechanics

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Abstract

We present a candidate mathematical framework for understanding coordination collapse in complex societies, integrating game theory, information theory, network science, statistical mechanics, evolutionary biology, and AI safety. Building on empirical observations from 48 historical collapse events, we propose **seventeen candidate regularities** governing trust dynamics, cascade acceleration, network topology effects, recovery probability, metacognitive failure, evolutionary stability, and AI-coordination interactions.

The framework **models** coordination collapse as a **candidate phase transition** with mean-field critical exponents ($\beta = 0.5$, $\gamma = 1.0$), derives the trust threshold ($\theta \approx 0.375$, model-dependent) from coordination game payoffs, and **proposes** that trust (H_3) functions as the **information hub** connecting all other coordination dimensions. These results are suggestive but require further empirical validation.

Key innovations include: (1) distinguishing centralization (cascade onset) from redundancy (cascade velocity), resolving the Rome Paradox; (2) formalizing “Dark Trust” as unmeasured coordination capacity; (3) deriving the threshold from coordination game payoffs; (4) modeling coordination collapse as a Landau-type phase transition; (5) extending the framework to Kardashev-scale civilizations; (6) **proposing K as metacognitive capacity**—civilizations collapse when they lose the ability to model themselves; (7) **showing that under baseline replicator dynamics (one-shot, well-mixed populations), cooperation is not an ESS**— θ marks the basin boundary requiring institutional maintenance; (8) **the AI Coordination Paradox**—AI amplifies coordination dynamics in both directions simultaneously, and may be a proximate mechanism of Great Filter events for technological civilizations.

The framework aims to unify the mathematics governing phase transitions, evolutionary dynamics, and information flow with civilizational collapse dynamics. The threshold $\theta \approx 0.375$ is consistent with multiple theoretical derivations (game theory, percolation theory, and information theory), though the precise value is model-dependent. The framework **proposes** a resolution to the Fermi Paradox: (1) the coordination threshold may function as a Great Filter mechanism; (2) survivors might maintain deliberate silence because contact could increase modernization () faster than trust develops; (3) AI may accelerate these dynamics. This generates falsifiable predictions including early warning signals (critical slowing down, increased variance), specific scaling relationships, and the hypothesis that humanity is currently in a critical coordination transition.

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1 Introduction

1.1 The Problem

Why do civilizations collapse? Despite millennia of historical examples and centuries of scholarly analysis, we lack a predictive framework that: - Quantifies when collapse becomes likely - Explains why some collapses are fast, others slow - Predicts whether recovery is possible - Guides intervention timing and design

1.2 Our Contribution

We present the **Coordination Collapse Framework**—a set of **five Core Propositions with moderate empirical support** (Props. 1, 2, 5, 9, 10), plus **seven Supporting Regularities** (Props. 3, 4, 6, 7, 8, 11, 12) and **five Theoretical Extensions** (Props. 13-17)—seventeen candidate relationships, motivated by game theory, network science, information theory, and statistical mechanics. The Core Propositions show consistency with historical data (N=48); the Extensions represent theoretical hypotheses requiring empirical testing.

This framework aims to: 1. **Propose** a unified vocabulary connecting social science with physics (phase transition formalism) 2. **Derive** (not just describe) candidate collapse dynamics from first principles 3. **Explore** scaling from human civilizations to interstellar scales (speculative) 4. **Generate** testable predictions applicable across historical cases

1.3 Paper Structure

- Section 2: The Core Engine (Laws 1, 2, 9, 10) — includes Core Model & Notation
- Section 3: The Network Architecture (Law 3)
- Section 4: The Dark Trust Framework (Law 8)
- Section 5: Dynamics and Implications (Laws 4-7, 11-12)
- **Section 6: Information-Theoretic Foundations (Law 13)**
- **Section 7: Phase Transition Framework (Law 14)**
- **Section 8: Metacognitive Collapse Theory (Law 15)**
- **Section 9: Evolutionary Stability Analysis (Law 16)**
- **Section 10: The AI Coordination Paradox (Law 17)**
- Section 11: Validation and Predictions
- Section 12: Kardashev Extension and the Great Filter
- **Section 13: Related Work [NEW]**
- Section 14: Discussion

1.4 2. The Core Engine: Threshold, Cascade, and Feedback

1.4.1 2.0 Core Model and Notation

Before presenting the laws, we establish the core model and notation used throughout this paper.

Core Variables: | Symbol | Definition | Units/Range | |——|———|———-| | H(t) | Trust/social cohesion dimension | [0, 1] | | K(t) | Coordination index (geometric mean of 7 harmonies) | [0, 1] | | | Critical trust threshold | 0.375 (model-dependent) | | | Modernization/cascade amplification factor | [1,) | | R | Coordination redundancy | [1,) | | C | Centralization degree | [0, 1] |

Terminology Note: This paper uses descriptive labels for harmonies suited to historical case analysis (e.g., H = “Trust/Social Cohesion”). Paper 1 (Historical K-Index) uses complementary aspirational terminology for modern contexts (e.g., H = “Cooperative Reciprocity,” aspirationally “Sacred Reciprocity”). The underlying constructs are equivalent; terminology differs to suit each paper’s analytical focus. See Paper 1 Table S1 for complete harmony definitions and data sources.

The Threshold Derivation (Model-Dependent):

The threshold emerges from a stylized N-player coordination game. Consider a population where each agent chooses to cooperate (C) or defect (D). Let: - B = benefit from mutual cooperation - c = cost of cooperation - p = fraction of cooperators in the population

An agent cooperates if expected payoff exceeds defection:

$$E[\text{Cooperate}] > E[\text{Defect}]$$

$$p \times B - c > 0$$

$$p > c/B$$

For typical values ($c/B = 0.375$), this yields 0.375 . **Important:** This threshold is model-dependent, not a fundamental constant: - Different payoff structures yield different thresholds (range: 0.25-0.50) - The empirical estimate $\text{emp} = 0.375 \pm 0.025$ is consistent with this range - The proximity to $1/2 = 0.5$ is explored in **Paper 2B** (see below)

Relationship to Paper 2B: The companion paper “The Golden Threshold” demonstrates that **nine theoretical frameworks** (sharing structural assumptions)—from game theory, percolation physics, bifurcation analysis, information theory, thermodynamics, evolutionary biology, network science, maximum entropy principles, and renormalization group methods—converge on 0.382 ± 0.004 . While these frameworks are not strictly independent (they share assumptions about binary cooperation, local connectivity, and positive feedback), their convergence to such a narrow band is suggestive of an underlying regularity. See Paper 2B for derivations and caveats.

Epistemic Classification (see Analysis Package for details): | Tier | Laws | Count | Evidence
 | |—|—|—|—|—| | Core Laws | 1, 2, 5, 9, 10 | 5 | Strong ($R^2 > 0.7$, LOOCV validated)
 | | Regularities | 3, 4, 6, 7, 8, 11, 12 | 7 | Moderate-Weak | | Extensions | 13, 14, 15, 16, 17 | 5 |
 Theoretical (requires validation) | | **Total** | | **17** | |

1.4.2 2.1 Law 1: The Threshold (0.375)

Statement: There exists a critical trust level below which defection becomes the Nash equilibrium.

Derivation:

$$\text{Cooperation payoff: } U_C = p \times B - (1-p) \times C$$

$$\text{Defection payoff: } U_D = 0$$

Cooperation dominates when:

$$p \times B - (1-p) \times C > 0$$

$$p > C / (B + C)$$

For typical $B = 1.0$, $C = 0.6$:

$$= 0.6 / 1.6 = 0.375$$

1.4.3 2.2 Law 2: The Cascade (Quadratic Acceleration)

Statement: Below , trust decline accelerates quadratically.

Derivation: From network amplification dynamics:

$$dH/dt = -(H) - (H)^2$$

The quadratic term arises from:

$$D_{\text{effective}} = D \times (1 + \times D)$$

Validation: $R^2 = 0.74$ (quadratic) vs 0.59 (linear) vs 0.69 (exponential)

1.4.4 2.3 Law 9: The Feedback Loop

Statement: Trust generates conditions for more trust; distrust generates conditions for more distrust.

Derivation: [Detailed in Appendix]

1.5 3. The Network Architecture: Resolving the Rome Paradox

1.5.1 3.1 The Paradox

Rome was hierarchical (Emperor-centered) yet collapsed slowly (250 years). Soviet was hierarchical (Party-centered) yet collapsed rapidly (6 years).

The naive prediction: Hierarchy $\rightarrow$ Fast collapse (single point of failure) The reality: Rome slow, Soviet fast.

1.5.2 3.2 Law 3: The Resolution

Key Insight: Distinguish Centralization (C) from Redundancy (R)

- **Centralization:** Affects cascade *onset* (can single shock trigger cascade?)
- **Redundancy:** Affects cascade *velocity* (how fast does cascade spread?)

Formula:

$$v_{\text{cascade}} \propto 1/R$$

where R = number of independent coordination mechanisms

Validation: | Case | C | R | Predicted Speed | Actual | |—|—|—|—|—|—|—|—|—| | Rome
| 0.7 | 3.3 | Slow | Slow | | Soviet | 0.9 | 1.6 | Fast | Fast |

1.5.3 3.3 Visual Representation

[Figure 2: Topology Comparison - Star vs Mesh vs Clustered]

151 1.6 4. The Dark Trust Framework: Explaining the Soviet Paradox

152 1.6.1 4.1 The Paradox

153 Soviet surveys showed low interpersonal trust (~ 0.20), yet the system functioned. How can $H <$
154 coexist with apparent coordination?

155 1.6.2 4.2 Law 8: Dark Trust

156 **Definition:** Total coordination capacity includes unmeasured components.

$$157 H_{\text{total}} = H_{\text{light}} + H_{\text{dark}}$$

158

159 **where:**

160 H_{light} = Survey-measurable organic trust

161 H_{dark} = H_{coerced} + H_{habitual} + H_{implicit}

162 **Soviet Example:** - $H_{\text{light}} = 0.20$ (surveys) - $H_{\text{coerced}} = 0.25$ (KGB-enforced compliance)
163 - $H_{\text{habitual}} = 0.10$ (behavioral inertia) - $H_{\text{total}} = 0.55$ (above , hence functional)

164 1.6.3 4.3 The Coercion Cliff

165 When enforcement fails, H_{coerced} collapses instantaneously:

$$166 d(H_{\text{coerced}})/dt = - \text{ at shock event}$$

167 This explains why authoritarian collapses are **discontinuous** (sudden) while democratic col-
168 lapses are **continuous** (gradual erosion).

169 [Figure 3: The Dark Trust Iceberg / Coercion Cliff]

170

171 1.7 5. Dynamics and Implications

172 1.7.1 5.1 Law 4: Modernization

173 **Statement:** Modern civilizations collapse faster due to increased information speed, economic
174 integration, and network connectivity.

175 **Formula:**

$$176 = _base \times F_{\text{info}} \times F_{\text{econ}} \times F_{\text{network}}$$

177 1.7.2 5.2 Law 5: Recovery

178 **Statement:** Recovery probability is $\sim 15\%$ below threshold, decaying $\sim 5\%$ /year.

$$179 P(\text{recovery} \mid H <) = 0.15 \times e^{(-0.05 \times \text{years_below})}$$

1.7.3 5.3 Law 7: Intervention Window

Statement: ROI is $\sim 10:1$ above threshold, $\sim 1:10$ below.

Derivation: Three asymmetry sources: 1. Incentive alignment ($3\times$) 2. Sustainability ($5\times$) 3. Social amplification ($2\times$)

Combined: $30\times$ difference at extremes

[Figure 4: Intervention ROI Curve]

1.7.4 5.4 Law 12: Glass Ceiling

Statement: $K_{\max} = 0.85$ due to five converging limits.

Connection to Ashby's Law: Perfect coordination ($K=1.0$) implies zero internal variety, preventing adaptation to environmental change.

1.8 6. Information-Theoretic Foundations (Law 13) [NEW]

1.8.1 6.1 Law 13: The Information Hub Law

Statement: Trust (H_3) is the “information hub” of civilizational coordination. The threshold $\theta = 0.375$ corresponds to the half-entropy point of the coordination state space.

1.8.2 6.2 Why H_3 Fails First: Information Structure (Hypothesis)

Proposition: H_3 carries high mutual information with other harmonies:

$I(H_3; H_1) = \text{high}$ [Governance]

$I(H_3; H_2) = \text{high}$ [Networks]

$I(H_3; H_4) = \text{moderate}$ [Complexity]

$I(H_3; H_5) = \text{moderate}$ [Knowledge]

$I(H_3; H_6) = \text{moderate}$ [Wellbeing]

$I(H_3; H_7) = \text{lower}$ [Technology]

Note: Specific values require empirical estimation from cross-national data. The qualitative ordering (H as hub) is hypothesized, not proven.

1.8.3 6.3 Threshold from Shannon Entropy

The coordination state entropy:

$S_{\text{coord}} = -\sum_p p(\text{config}) * \log(p(\text{config}))$

Setting $S_{\text{coord}}(\theta) = S_{\max}/2$ (half-entropy point):

$\theta = 0.382$

This matches the game-theoretic derivation AND the golden ratio ($1/\phi^2$)!

215 1.8.4 6.4 Information Cascade Dynamics

$$216 \quad dI/dt = -\lambda * (\theta - H_3)^2 * I(H_3; H_{rest})$$

217

218 Information loss accelerates quadratically below threshold.

219 [Figure 10: Information Entropy Landscape]

220

221 1.9 7. Phase Transition Framework (Law 14) [NEW]

222 1.9.1 7.1 Law 14: Coordination Collapse as Phase Transition

223 **Statement:** Civilizational coordination collapse is a second-order phase transition, analogous to
224 ferromagnetic transitions, with universal critical exponents.

225 1.9.2 7.2 The Landau Free Energy

$$226 \quad F(m, T) = a(T)m^2 + b m^4$$

227

228 Where:

- 229 - $m = K - K_c$ (order parameter)
- 230 - $a(T) = a_0(T - T_c)/T_c$ (changes sign at transition)
- 231 - T_c corresponds to $H_3 = \theta$

232

233 Above θ : Single minimum (cooperation stable)

234 Below θ : Double well (defection favored)

235 1.9.3 7.3 Proposed Critical Exponents (Mean-Field Hypothesis)

	Exponent	Value	Physical Meaning
$\beta = 0.5$	$m \sim \theta - H_3 ^{0.5}$		Order parameter scaling
$\gamma = 1.0$	$\chi \sim \theta - H_3 ^{-1}$		Susceptibility divergence
$\nu = 0.5$	$\xi \sim \theta - H_3 ^{-0.5}$		Correlation length

236 **Note:** These are mean-field exponents, predicted IF the phase transition analogy holds AND
237 social networks approximate mean-field conditions (high connectivity, weak local clustering). Actual
238 social networks may exhibit different exponents due to: - Strong local correlations (community
239 structure) - Long-range connections (small-world properties) - Heterogeneous degree distributions
240 (power-law networks) - Agency and adaptation (agents respond to their environment)

241 **Empirical validation of these exponents in social systems remains an open challenge.**

242 1.9.4 7.4 Early Warning Signals

243 Before collapse, the system shows: 1. **Critical slowing down:** $\tau \sim 1/|H_3 - \theta|$ (recovery
244 takes longer) 2. **Increased variance:** $\text{Var}(K) \sim |H_3 - \theta|^{-1}$ 3. **Increased autocorrelation:**
245 Approaching $C(t) \rightarrow 1$

246 These signals are GENERIC indicators of approaching phase transition.

247 [Figure 8: Landau Free Energy Landscape] [Figure 9: Critical Exponent Scaling]

1.10 8. Metacognitive Collapse Theory (Law 15) [NEW]

1.10.1 8.1 Law 15: The Metacognitive Collapse Law

Statement: Civilizational collapse is fundamentally a failure of metacognition—the civilization’s ability to model and predict its own behavior.

$M = K \times R_{\text{model}} \times T_{\text{response}}$ (Metacognitive capacity)

Where:

- K = Coordination index
- R_{model} = Model accuracy (prediction-outcome correlation)
- T_{response} = Response time

Collapse condition: $M < C_{\text{env}}$ (environmental complexity)

1.10.2 8.2 K as Civilizational Self-Awareness

Harmony	Metacognitive Interpretation
H (Governance)	Collective intention formation
H (Networks)	Information integration
H (Trust)	Shared world model
H (Complexity)	Response repertoire
H (Knowledge)	Predictive capability
H (Wellbeing)	Coherence signal
H (Technology)	Action capability

Trust (H) is central because it enables SHARED WORLD MODELS.

1.10.3 8.3 The Blindness Before Collapse

Empirical observation: Collapsing civilizations fail to see collapse coming. - Rome 400 CE: No contemporaneous writings predicted the end - Soviet 1989: Western experts predicted decades of stability - Maya 800 CE: No recorded warnings

Explanation: When $M < C_{\text{env}}$, the civilization CANNOT MODEL ITS OWN DYNAMICS.

$P(\text{accurate self-prediction} \mid M < C_{\text{env}}) \rightarrow 0$

The civilization becomes BLIND to itself.

1.10.4 8.4 The Three Levels of Civilizational Consciousness

```

    ( ) ( ) ( )
* * * *
0.1079330125K4Principles
0.2308Range


---


    ( ) ( ) ( )
* * * *
0.1079330125K4None
3: dict's achieved
And 0.75 per-
ticpre- ma-
i- ventsnently
paerises
tory
    ( ) ( ) ( )
* * * *
0.1079330125K4Most
2: sponsa-
Reto K ble
ac-crisestates
tivef-0.75
    ter
    on-
    set
    ( ) ( ) ( )
* * * *
0.1079330125K4Pre-
1: not < collapse
Blind 0.45v-
    own i-
    dy- liza-
    nam-tions
    ics
    ( ) ( ) ( )
* * * *
0.1079330125K4Failed
0: co- < states
Collet 0.30
lapsed
    self-
    model

```

272 1.10.5 8.5 The Meta-Paradox

273 If metacognition is failing, CAN IT RECOGNIZE that it's failing?

274 M_meta = ability to model M

275

276 If $M < C_{env}$, then M_meta is also compromised.

277

278 META-BLINDNESS: The civilization can't see that it can't see.

279

This explains why external observers often see collapse before internal actors.

280

[Figure 13: Metacognitive Levels and Collapse Trajectory]

281

282 1.11 9. Evolutionary Stability Analysis (Law 16) [NEW]

283 1.11.1 9.1 Law 16: The Fragile Cooperation Law

284 **Statement:** Under baseline replicator dynamics (one-shot, well-mixed populations without insti-
285 tutions), cooperation is NOT an Evolutionarily Stable Strategy (ESS). The threshold 0.375 marks
286 the boundary of the basin of attraction for cooperative equilibria.

287 **Scope Limitation:** This applies specifically to: - One-shot games (no repeated interaction) -
288 Well-mixed populations (no spatial/network structure) - No institutional enforcement mechanisms

289 This does NOT contradict established results on cooperation via: - Network reciprocity (Nowak
290 & May, 1992) - Repeated games (Axelrod, 1984) - Group selection mechanisms (Boyd & Richerson,
291 1985) - Institutional design (Ostrom, 1990)

292 1.11.2 9.2 Why Cooperation is Unstable Under Baseline Conditions

293 From standard replicator dynamics in one-shot games:

$$\begin{aligned} 294 \quad dp/dt &= p(1-p) [E(\text{Cooperate}) - E(\text{Defect})] \\ 295 \quad &= p(1-p)(-c) \\ 296 \quad &= -cp(1-p) \end{aligned}$$

297

298 Where p = fraction of cooperators, c = cost of cooperation

299

300 Since $c > 0$: $dp/dt < 0$ for all $0 < p < 1$

301

302 Under these specific conditions, cooperation declines.

303 **Key insight:** Defection is the ESS under baseline conditions. Cooperation requires active
304 institutional maintenance to override evolutionary defaults.

305 1.11.3 9.3 The Three Equilibria

	() () () ()	
* * * *		
0.3250	0.3250	0.3250
	() () () ()	
* * * *		
0.3250	0.3250	0.3250
De- blai-		
fec- ESSal		
tion	evo-	
	lu-	
	tion-	
	ary	
	out-	
	come	
	() () () ()	
* * * *		
0.3250	0.3250	0.3250
0.3750	0.3750	0.3750
	blary	
	() () () ()	
* * * *		
0.3250	0.3250	0.3250
	staples	
	con-	
	tin-	
	u-	
	ous	
	main-	
	te-	
	nance	

306 Below : System flows toward $p = 0$ (collapse inevitable)
307 Above : Cooperation can be maintained with institutions
308 At : Bifurcation point-qualitative phase transition

309 **1.11.4 9.4 The Cost of Civilization**

310 Cost_civilization = Maintenance(t) dt
311

312 Civilization is a "dissipative structure":

- 313 - Requires continuous energy input
- 314 - Maintains low-entropy (ordered) state
- 315 - Stops paying $\rightarrow$ returns to high-entropy (defection)

316 1.11.5 9.5 Connection to Thermodynamics

317 Cooperation = Low entropy state (ordered)

318 Defection = High entropy state (disordered)
 319
 320 Second Law: Entropy increases spontaneously.
 321 Civilization requires WORK to maintain order.

322 1.11.6 9.6 The Deep Insight

323 Natural selection produces defection.
 324 Cooperation requires CULTURAL EVOLUTION to override.
 325
 326 Civilization = Cultural override of biological default.
 327 Collapse = Return to evolutionary default.

328 **The threshold 0.375 is WHERE CULTURAL SELECTION LOSES TO BIOLOG-**
 329 **ICAL SELECTION.**

330 [Figure 14: Evolutionary Stability Landscape]

331

332 1.12 10. The AI Coordination Paradox (Law 17) [NEW]

333 1.12.1 10.1 Law 17: The AI Coordination Law

334 **Statement:** Artificial intelligence amplifies the coordination dynamics by factors $A_{\text{pos}}(t)$ and
 335 $A_{\text{neg}}(t)$, where:

336 $dH/dt_{\text{with_AI}} = dH/dt_{\text{natural}} \times (1 + A_{\text{pos}}(t))$
 337 $d/dt_{\text{with_AI}} = d/dt_{\text{natural}} \times (1 + A_{\text{neg}}(t))$

338

339 If $A_{\text{neg}} > A_{\text{pos}}$: Civilization accelerates toward threshold.

340 If $A_{\text{pos}} > A_{\text{neg}}$: Civilization accelerates toward safety.

341

342 The paradox: Both effects come from the SAME technology.

343 **Connection to Technology Evaluation Framework:**

344 A_{pos} and A_{neg} connect directly to the H and effects from the Coordination Technology
 345 Framework:

346 $A_{\text{pos}}(t) = H(\text{AI_deployment}) / H_{\text{baseline}}$

347 $A_{\text{neg}}(t) = (\text{AI_deployment}) / _baseline$

348

349 Technology Safety Score = $H - \times$ (where 0.5)

350

351 If Score > 0: Technology net-positive for coordination

352 If Score < 0: Technology net-negative for coordination

353 This provides an operationalizable framework for AI safety: assess whether a given AI system
 354 increases H (trust, verification, translation) more than it increases (cascade velocity, modernization
 355 pressure).

356 **AI is the first technology that can change faster than any natural H adaptation.**

357 1.12.2 10.2 Why AI is Unique Among Technologies

358 Unlike previous technologies:

	Property	Previous Tech	AI
Speed	Gradual adoption		Exponential deployment
Scope	Domain-specific		Universal capability
Autonomy	Human-operated		Self-directed action
Trust impact	Local		Global (simultaneously)
Reversibility	Possible		Path-dependent

359 1.12.3 10.3 Three Qualitative AI Scenarios

360 **Note:** The following scenarios are qualitative descriptions, not calibrated probability estimates.
 361 We do not have sufficient empirical grounding to assign meaningful probabilities.

362 **Scenario A: Coordination Collapse (possible)**

- 363 - AI accelerates faster than H can adapt
- 364 - Economic disruption, misinformation, polarization
- 365 - H crosses → Irreversible cascade

367 **Scenario B: Coordination Enhancement (possible)**

- 368 - AI governance established early
- 369 - AI used for verification, translation, coordination
- 370 - H rises above + margin → Sustainable civilization

372 **Scenario C: Unstable Equilibrium (possible)**

- 373 - AI effects roughly balance
- 374 - Repeated near-threshold crises
- 375 - Eventually resolves into A or B

376 Which scenario unfolds depends on policy choices, technological development paths, and social
 377 dynamics that are difficult to predict.

378 1.12.4 10.4 AI as Great Filter Mechanism

379 **Hypothesis:** AI may be the PROXIMATE CAUSE of most Great Filter events.

380 $_AI$ = Time to develop transformative AI

381 $_$ = Time to reliably maintain H >

383 If $_AI < _$ (typical case):

- 384 AI arrives before coordination is solved.
- 385 High collapse probability.

387 If $_ < _AI$ (rare):

- 388 Coordination is stable before AI arrives.
- 389 AI can be deployed safely.
- 390 Civilization likely survives to Type II.

391 **Earth's situation:** $_AI$ appears to be NOW. $_$ has not been reached.

392 $P(\text{collapse} \mid AI \text{ before solved}) \gg P(\text{collapse} \mid \text{solved before AI})$

393

394 Most civilizations develop AI before solving coordination physics.

395 AI may be the mechanism by which most Great Filter events occur.

396 1.12.5 10.5 The Alignment Problem as Coordination Physics

397 Coordination physics reframes AI alignment:

398 Traditional: "How do we ensure AI does what humans want?"

399 Coordination: "How do we ensure AI maintains $H > ?$ "

400

401 $\text{Alignment_coordination} = P(H > \mid AI_deployed)$

402

403 An aligned AI is one that:

404 1. Does not accelerate faster than H can adapt

405 2. Actively supports trust-building processes

406 3. Maintains human agency in coordination

407 4. Preserves shared reality and common ground

408 **True alignment = AI that makes civilization MORE STABLE, not just more capa-**
409 **ble.**

410 1.12.6 10.6 Current Status Assessment

411 **Note:** The following is a rough estimate with significant uncertainty. H measurements are imprecise,
412 and the threshold is model-dependent.

413 Estimated Earth status (2024-2025):

414 H 0.40-0.50 (uncertain, varies by measurement)

415 0.35-0.40 (model-dependent)

416 Margin: unclear due to measurement uncertainty

417

418 If the framework is correct, current conditions merit attention

419 but do not support precise timeline predictions.

420 1.12.7 10.7 The Deep Truth

421 AI is a mirror.

422

423 High H civilization + AI = Enhanced coordination

424 Low H civilization + AI = Accelerated collapse

425

426 AI does not determine our fate.

427 It AMPLIFIES the fate we are already choosing.

428

429 We are currently in the middle zone.

430 Everything depends on choices made NOW.

1.13 11. Validation and Predictions

1.13.1 11.1 Hindcast Validation

Table S1: Standardized Dataset

1.13.2 11.2 Sensitivity Analysis

- Rankings stable under $\pm 20\%$ parameter perturbation
- Rome always slower than Soviet in 100% of scenarios
- H_3 (Trust) is dominant driver: -96% to +224% sensitivity

1.13.3 11.3 Model Validity Assessment

Prediction
Validation
Type
Rankings
Soviet
in
100%
of
sce-
nar-
ios
Relative
Dy-
name
ics
con-
firmed
(R^2
=
0.74)

452

453 Type I ($E \sim 10^{17}$ W): $K_{\max} \sim 0.91$
454 Type II ($E \sim 4 \times 10^{26}$ W): $K_{\max} \sim 0.96$
455 Type III ($E \sim 4 \times 10^{37}$ W): $K_{\max} \sim 0.99$

456 1.14.2 12.2 Five Universal Limits Constraining $K_{\max}$

- 457 1. **Ashby's Limit:** Perfect coordination ($K=1$) = zero variety = no adaptation
- 458 2. **Dunbar's Limit:** Cognitive constraints on coordination relationships
- 459 3. **Shannon's Limit:** Channel capacity bounds information sharing
- 460 4. **Light-Speed Limit:** Coordination delay grows with spatial extent
- 461 5. **Entropy Production:** Second Law requires dissipation for coordination

462 1.14.3 12.3 The Great Filter as Coordination Threshold

463 **Hypothesis:** The Great Filter in the Fermi Paradox may be related to the coordination threshold
464 .

465 **Why Civilizations Fail to Reach Type I** For a civilization to transition from Type 0 (plan-
466 etary) to Type I (stellar), it must: 1. **Global coordination:** Climate action, resource sharing,
467 technology governance 2. **Multi-generational planning:** Century-scale projects 3. **Trust at**
468 **unprecedented scale:** Billions of agents coordinating

469 **The threshold 0.375 represents the minimum trust for this coordination.**

470 From our historical analysis:

471 $P(\text{recovery} \mid H <) = 0.15$

472

473 85% of civilizations that cross below threshold COLLAPSE.

474 **The Timeline Squeeze** The critical Type I transition period is ~ 100 -500 years. But moderniza-
475 tion () INCREASES collapse speed:

476 $v_{\text{cascade}} \propto (H - H)^2$

477

478 As technology advances, collapse accelerates when trust drops.

479 This creates a narrowing window for Type I achievement.

480 **Survival Probability (Illustrative Calculation)** **Note:** The following is a rough illustrative
481 calculation, not a calibrated probability estimate. All parameter values are uncertain and the model
482 structure itself is unvalidated.

483 $P(\text{Type I}) = P(H > \text{throughout}) + P(\text{recovery} \mid \text{crossed}) \times P(\text{cross})$

484

485 Assumed parameters (illustrative only):

- 486 - $H_0 = 0.50$ (post-agricultural trust, uncertain)
- 487 - $\dot{H} = 0.02/\text{year}$ (volatility, rough estimate)
- 488 - 300 years (transition period, model-dependent)
- 489 - 0.375 (threshold, model-dependent)

490

491 With these assumptions:

492 $P(\text{Type I}) \sim 0.3\text{--}0.7$ (order-of-magnitude estimate only)

493

494 Similar uncertainty applies to:

495 $P(\text{Type II} \mid \text{Type I}) \sim \text{uncertain}$

496 $P(\text{Type II} \mid \text{intelligence}) \sim \text{highly uncertain}$

497 These numbers are meant to illustrate the framework's structure, not to provide
498 reliable forecasts.

499 Modified Drake Equation (Conceptual)

500 $N = R^* \times f_p \times n_e \times f_l \times f_i \times f_c \times f_{\text{coord}} \times L$

501

502 Where $f_{\text{coord}} = P(\text{Type II} \mid \text{intelligence}) = \text{unknown}$

503

504 If this framework is correct, f_{coord} could be small (<0.5),
505 potentially reducing detectable civilizations significantly.

506 **Note:** We cannot reliably estimate f_{coord} from our framework. The point is conceptual: IF
507 coordination thresholds exist universally, they may contribute to the Great Filter.

508 Testable Predictions

- 509 1. **Historical (validated):** $\sim 15\%$ recovery rate below threshold (observed: $15\% \pm 5\%$)
- 510 2. **Contemporary:** Civilizations near show increased variance, slower crisis recovery
- 511 3. **SETI:** Detected signals should come from stable ($H \gg$), ancient civilizations
- 512 4. **Great Timing:** We exist during the rare Type I transition window (10^{-8} of existence)

513 **Speculation:** If coordination thresholds are universal, the silence of the cosmos could partially
514 reflect coordination failures. This is one of many possible contributions to the Fermi Paradox, not
515 a definitive explanation.

516 [Figure 12: The Great Filter as Coordination Threshold]

517 1.14.4 12.4 The Wisdom Silence Hypothesis: Why Survivors Don't Contact Us

518 **Revolutionary Extension:** Civilizations that survive to Type II+ necessarily understand coordi-
519 nation physics. This knowledge implies they would NOT contact pre-threshold civilizations.

520 **The Contact Danger** Technology transfer accelerates modernization () faster than trust can
521 develop:

522 $d/dt_{\text{contact}} \gg dH/dt_{\text{natural}}$

523

524 Even modest technology sharing ($2\times$ amplification):

525 - Doubles cascade velocity if threshold crossed

526 - Halves adaptation time

527 - Transforms manageable decline into catastrophe

528 **The Wisdom Silence** Advanced civilizations maintain **deliberate silence**—not from fear (Dark
529 Forest) but from **compassion**:

	What They Could Share	Risk	Reason
Virtues/Ethics	Low	Builds H, doesn't increase	
Coordination wisdom	Low	Teaches threshold management	
Applied technology	High	Rapid increase, trust gap	
Energy technology	Very High	Massive amplification	

530 **The only safe gifts are WISDOM gifts—not technology gifts.**

531 **The Coordination Quarantine**

532 **Graduation_criteria:**

- 533 1. $H > +$ margin for _proof > 200 years
- 534 2. Demonstrated planetary-scale coordination
- 535 3. Evidence of understanding coordination physics

536
537 **Earth status:** FAILS ($H = 0.42$, declining, near threshold)

538 **Implication (Speculative)** **Note:** This section is highly speculative. We have no evidence of
539 advanced civilizations, let alone their motivations.

540 IF advanced civilizations exist AND they understand coordination dynamics, one possible reason
541 for non-contact would be concern about destabilizing less-developed civilizations through technol-
542 ogy transfer. This is one of many possible explanations for the Fermi Paradox, not a confirmed
543 hypothesis.

544 [Figure 15: The Wisdom Silence - Contact Safety Boundary]

545 **1.14.5 12.5 Future Research Directions**

- 546 1. **Experimental validation:** Laboratory studies of coordination games
- 547 2. **Neural network monitoring:** Real-time early warning systems
- 548 3. **Multi-civilization dynamics:** Interaction between societies
- 549 4. **Quantum effects:** Fundamental limits on coordination at extreme scales

550 [Figure 11: Kardashev Scaling of K_{max}]

551

552 **1.15 13. Related Work**

553 This framework builds on and extends several established research traditions. We situate each major
554 component in the existing literature:

1.15.1 13.1 Collapse Theory

Complexity-based approaches: Tainter (1988) introduced the concept of diminishing returns on complexity investment. Our framework incorporates this through the Glass Ceiling (Law 12) and the modernization factor , but adds the crucial distinction that complexity is not the root cause—trust erosion is.

Environmental factors: Diamond (2005) emphasizes environmental degradation and failure to adapt. We treat these as H (wellbeing) and H (technology) effects that manifest *through* trust erosion, rather than independently causing collapse.

Secular cycles: Turchin (2003, 2023) models elite overproduction and popular immiseration as drivers of instability cycles. Our cascade dynamics (Law 2) can be understood as the mechanism underlying these cycles—when elite competition erodes trust, the cascade accelerates.

War and state formation: Scheidel (2017) documents how violence shapes inequality and institutions. Our Dark Trust framework (Law 8) captures how coercion-based coordination differs from organic trust.

1.15.2 13.2 Evolutionary Game Theory

Cooperation stability: Nowak (2006) identifies five mechanisms enabling cooperation (kin selection, direct reciprocity, indirect reciprocity, spatial structure, group selection). Law 16’s scope limitation explicitly acknowledges these—we claim only that *baseline* conditions favor defection, not that cooperation is impossible.

Institutional design: Ostrom (1990) demonstrates how communities solve collective action problems through institutional design. This informs our emphasis on institutional maintenance above the threshold.

1.15.3 13.3 Phase Transitions and Critical Phenomena

Social phase transitions: Scheffer et al. (2009) apply critical transition theory to social-ecological systems, identifying early warning signals. Law 14 extends this by proposing specific critical exponents (mean-field) and connecting to coordination game theory.

Network percolation: The connection between and percolation threshold (Law 10) draws on standard network science (Barabási, 2016; Newman, 2018).

1.15.4 13.4 AI Safety

Coordination as alignment: Russell (2019) and Bostrom (2014) frame AI safety in terms of value alignment. Our Law 17 reframes this: alignment is fundamentally about maintaining $H > 0$, i.e., ensuring AI supports rather than undermines coordination capacity.

Existential risk: Ord (2020) estimates existential risk from AI at $\sim 10\%$ this century. Our framework provides a mechanistic explanation: AI accelerates faster than H can adapt, potentially triggering coordination collapse.

1.15.5 13.5 What This Framework Adds

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592 1.16 14. Discussion

593 1.16.1 14.1 Implications for Policy

- 594 1. **Prevention » Cure:** ROI asymmetry demands pre-threshold intervention
- 595 2. **Build Redundancy:** More coordination mechanisms = slower collapse
- 596 3. **Convert Dark Trust:** Transition from coerced to organic trust
- 597 4. **Monitor Early Warning:** Track variance, autocorrelation, recovery time

598 1.16.2 14.2 Contributions and Scope

599 This framework attempts to: - **Integrate** concepts from game theory, network science, information
600 theory, statistical mechanics, and AI safety into a common vocabulary - **Propose** (not definitively
601 derive) mechanisms for collapse dynamics - **Generate** testable predictions based on historical pat-
602 terns (N=48) - **Extend** (speculatively) to larger scales, pending validation - **Suggest** connections
603 between social science and physics through phase transition analogies - **Reframe** AI alignment as
604 a coordination problem (a perspective, not a proven equivalence)

605 1.16.3 14.3 A Proposed Interpretive Framework

606 **Traditional approach:** Collapse explained through case-specific historical factors (war, climate,
607 complexity)

608 **Proposed framework:** Collapse modeled as coordination failure with candidate regularities:
609 1. Trust threshold () suggested by game theory models 2. Cascade dynamics hypothesized from
610 network effects 3. Phase transition analogy from statistical mechanics 4. Scaling extensions (spec-
611 ulative, requiring validation) 5. AI as amplifier of coordination dynamics (hypothesis)

612 **Note:** This framework proposes a unifying lens for understanding collapse, not a “physics of
613 civilization” in the literal sense. The analogy to physical phase transitions is suggestive and may

aid prediction, but social systems differ fundamentally from physical systems in agency, reflexivity,
and complexity. Treating these propositions as equivalent to physical laws would be an overreach.

1.16.4 14.4 Limitations

- Timing predictions require case-specific calibration
- Small sample size for recovery estimates (N=35)
- Critical exponents assumed mean-field (may vary by network topology)
- Threshold () may have cultural variation (~10%)
- AI scenario probabilities are estimates requiring validation

1.17 Figures

1.17.1 Main Text Figures (Core Framework)

1. **Figure 1:** The Core Engine (Threshold + Cascade Phase Diagram)
2. **Figure 2:** Topology Comparison (Soviet Star vs Market Mesh vs Rome Clustered)
3. **Figure 3:** Dark Trust Iceberg / Coercion Cliff
4. **Figure 4:** Intervention ROI Asymmetry Curve
5. **Figure 5:** Cascade Velocity Tornado (Sensitivity Analysis)
6. **Figure 6:** Ranking Stability Heatmap (Rome vs Soviet)
7. **Figure 7:** Cascade Velocity Phase Diagram

1.17.2 Extended Framework Figures (Laws 13-17)

8. **Figure 8:** Landau Free Energy Landscape (Phase Transition)
9. **Figure 9:** Critical Exponent Scaling (Universal Behavior)
10. **Figure 10:** Information Entropy Landscape (Law 13)
11. **Figure 11:** Kardashev Scaling of $K_{\max}$
12. **Figure 12:** The Great Filter as Coordination Threshold
13. **Figure 13:** Metacognitive Levels and Collapse Trajectory (Law 15)
14. **Figure 14:** Evolutionary Stability Landscape (Law 16)
15. **Figure 15:** The Wisdom Silence - Contact Safety Boundary
16. **Figure 16:** The AI Coordination Paradox - Amplification Dynamics (Law 17)

1.18 Supplementary Information

1.18.1 Table S1: Standardized Dataset

1.19.3 Network Science and Phase Transitions

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1.19.4 AI Safety and Existential Risk

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- Russell, S. (2019). *Human Compatible: Artificial Intelligence and the Problem of Control*. Viking.
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1.19.5 Additional References

- Schelling, T.C. (1960). *The Strategy of Conflict*. Harvard University Press.
- North, D.C. (1990). *Institutions, Institutional Change and Economic Performance*. Cambridge University Press.

1.20 Author Contributions

Tristan Stoltz: Conception, theoretical framework, historical interpretation, validation design
Claude (Anthropic): Mathematical formalization, derivation assistance, code generation, literature synthesis

This paper represents a synthesis of empirical observation, game-theoretic reasoning, and network science to create a predictive framework for civilizational coordination dynamics.